

# BPCL – IIM Visakhapatnam National Techno-Managerial Ideathon & Leadership Conference 2026–27

## *"Atmanirbhar Urja: Forging India's Sovereign Energy Future"*

**Organized by:** IIM Visakhapatnam

**Title Sponsor:** Bharat Petroleum Corporation Limited (BPCL) – CSR Initiative

**Target Audience:** Students of IITs, NITs, IIITs and other CFTIs/INIs

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### Objectives

- Identify and nurture future techno-managerial leaders for India's energy sector.
- Strengthen BPCL's positioning in energy transition and innovation.
- Build a high-quality talent pipeline and enhance awareness of IIMV's PGPMCI programme.
- Engage students from 120+ premier engineering institutions across the country.

### Ideathon Themes

#### **Theme 1: Hard-to-Abate Decarbonization — Solving India's Industrial Emissions**

*Innovations to decarbonize steel, cement, fertilizers, and refining — India's industrial backbone*

##### **Why This Theme Matters**

Steel, cement, fertilizers, and refining are the pillars of India's economic growth. Together, they contribute nearly 40% of India's industrial emissions. Unlike power generation, these sectors cannot simply switch to renewables — their emissions are intrinsic to their chemical processes. Decarbonizing them is India's toughest climate challenge and a critical step toward energy sovereignty. This theme invites participants to reimagine these foundational industries through breakthrough technologies and viable business models.

##### **Example Problem Areas**

- Techno-economic models for green hydrogen blending in steel manufacturing and fertilizer production
- Carbon capture and utilization (CCU) — converting captured CO<sub>2</sub> into building materials, soda ash, or synthetic fuels
- Alternative low-carbon fuels for cement kilns (bio-CNG, green hydrogen, biomass)
- Waste heat recovery and circular energy systems for refineries and industrial clusters
- Business models to make "green steel", "green cement", and "green fertilizers" commercially viable in India
- Policy and incentive frameworks to accelerate industrial decarbonization

**Techno-Managerial Balance:** 60% Technology / 40% Management

#### **Theme 2: Energy Storage & Grid Resilience — Powering India 24/7**

*Scalable solutions for storage, grid stability, and round-the-clock renewable power*

##### **Why This Theme Matters**

India has ambitious renewable energy targets, but solar and wind are intermittent. Without affordable, scalable energy storage, renewables cannot replace coal for base-load power. This is India's single biggest technical and economic bottleneck in the energy transition — solving storage will determine whether India achieves energy independence or remains dependent on fossil fuels. This theme challenges participants to think beyond generation and solve the real problem: making clean power reliable, dispatchable, and available 24/7.

#### Example Problem Areas

- Alternative battery chemistries suited to Indian conditions (sodium-ion, zinc-air, iron-flow, aluminium-air)
- Business models for utility-scale battery storage — revenue stacking, arbitrage, and ancillary services
- Pumped hydro storage — site identification, techno-economic feasibility, and environmental impact
- Vehicle-to-Grid (V2G) — business models enabling EV owners to sell stored power back to the grid
- AI/ML models for forecasting renewable energy output and optimizing grid dispatch
- Decentralized microgrids with storage for rural, island, and off-grid communities
- Hybrid renewable-plus-storage projects for round-the-clock (RTC) power supply

**Techno-Managerial Balance:** 50% Technology / 50% Management

### Theme 3: Critical Minerals & Circular Economy — Securing India's Raw Material Future

*Innovations in sourcing, recycling, and supply chain resilience for energy transition materials*

#### Why This Theme Matters

India imports 100% of its lithium, 70% of its cobalt, and 80% of its nickel. These minerals are the next oil — and India risks repeating the same dependency trap that has historically compromised its energy security. Without securing these raw materials, India's EV, battery, and renewable energy ambitions will remain vulnerable to global supply shocks and geopolitical tensions. Largely unexplored in student competitions, this theme offers fresh ground for breakthrough ideas that blend material science, supply chain strategy, and circular economy principles.

#### Example Problem Areas

- Urban mining models — recovering lithium, cobalt, copper, and rare earths from e-waste and end-of-life EV batteries
- AI-driven supply chain risk dashboards for real-time monitoring of critical mineral availability
- Business models for battery recycling, refurbishment, and second-life applications
- Alternative materials — reducing cobalt dependency in batteries or replacing rare earths in electric motors
- Strategic stockpiling — optimal sizing, location, and management of India's critical mineral reserves
- International partnership models for win-win sourcing from resource-rich nations (Australia, Chile, Africa, and others)
- Circular economy solutions for end-of-life solar panels and wind turbine blades
- Policy frameworks incentivizing domestic processing and recycling of critical minerals

**Techno-Managerial Balance:** 40% Technology / 60% Management

### Competition Structure

## Phase 1: National Outreach & Registration

- Target: 500+ teams (1,000+ students)
- One-slide idea submission
- Digital campaigns across 120+ CFTIs

## Phase 2: Evaluation & Shortlisting

- Top 90 teams shortlisted
- Detailed pitch deck and video submission
- Mentorship webinar with IIMV faculty and BPCL leaders

## Phase 3: Grand Finale at IIM Visakhapatnam

- 30 finalist teams (60 students)
- One-day event with keynote sessions, mentoring, and presentations
- Industry jury and BPCL leadership participation

## Evaluation Criteria

Submissions will be evaluated on the following parameters:

| Criteria                 | Weightage | Description                                                                                         |
|--------------------------|-----------|-----------------------------------------------------------------------------------------------------|
| Innovation & Originality | 30%       | Novelty of the idea — does it offer a fresh, non-obvious solution?                                  |
| Impact & Scalability     | 25%       | What is the potential environmental, economic, and strategic impact? Can it be scaled across India? |
| Technical Feasibility    | 20%       | Is the solution technically sound and implementable with current or near-future technology?         |
| Business Viability       | 15%       | Does the proposal have a clear, sustainable business model and a path to commercial adoption?       |
| Presentation & Clarity   | 10%       | How clearly is the idea communicated? Is the pitch compelling and well-structured?                  |

## Prize Pool — ₹1,80,000

| Category                              | Amount           |
|---------------------------------------|------------------|
| Overall Grand Winner                  | ₹75,000          |
| Three Theme Winners (₹25,000 each)    | ₹75,000          |
| Three Theme Runners-Up (₹10,000 each) | ₹30,000          |
| <b>Total Prize Pool</b>               | <b>₹1,80,000</b> |

## Value to BPCL

- Engagement with 1,000+ engineering students from premier institutions.
- Brand visibility among future leaders and innovators.
- Access to high-potential ideas aligned with BPCL's strategic priorities.
- Strengthening of BPCL's CSR and innovation ecosystem.

## Value to IIM Visakhapatnam

- Creation of a strong PGPMCI lead pipeline.
- Enhanced national visibility among CFTI students.
- Platform for academia–industry collaboration.

## Expected Outcomes

- 500+ team registrations from 75+ of the 120+ institutions reached
- 1,000+ students engaged nationwide
- 90 shortlisted ideas and 30 finalist teams
- National-level visibility for BPCL and IIM Visakhapatnam
- Long-term annual flagship event with scope for scale-up

## Road Map

| Milestone                                  | Timeline                |
|--------------------------------------------|-------------------------|
| Announcement                               | July 2026               |
| Phase 1: National Outreach & Registration  | August – September 2026 |
| Phase 2: Evaluation & Shortlisting         | October – November 2026 |
| Phase 3: Grand Finale at IIM Visakhapatnam | 23 January 2027         |

## Detailed Cash Budget

*Funding: BPCL ₹2.5 lakh (CSR) + PGPMCI admission outreach budget ₹2.5 lakh\**

| Head                                                                                       | Amount (₹) |
|--------------------------------------------------------------------------------------------|------------|
| Prizes                                                                                     | 1,80,000   |
| Digital campaigns across CFTIs (LinkedIn, Instagram, Google ads; WhatsApp broadcast tools) | 50,000     |
| Student travel support (₹1,000 × 60 students)                                              | 60,000     |
| Guest travel (3–4 speakers — train for three, one flight)                                  | 50,000     |
| Printing expenses (posters, banners, on-site branding, certificates, etc.)                 | 50,000     |
| Food for finale (80 persons × 1.5 days × ₹600)                                             | 80,000     |
| Miscellaneous / contingency                                                                | 30,000     |

| Head       | Amount (₹) |
|------------|------------|
| Total Cash | 5,00,000   |

### In-Kind Support from IIMV (Additional, Not Billed)

- Accommodation — 60 students + 15 guests in the guest house, one night
- Venue & AV — auditorium, three breakout rooms, projectors, and sound
- Faculty evaluation time — 6–8 faculty for screening and judging
- Staff coordination time — event manager and admin support

*(\*) Under the guidance of the PGPMCI Office, the student organising team will actively pursue additional funding for the ideathon/conference. This will help reduce the budgetary burden on the admissions outreach allocation as well as lessen the dependency on in-kind support from IIM Visakhapatnam. Furthermore, the additional funds will enable us to invite a larger number of industry participants to the conference. Targeted Sponsorship Categories:*

- *Gold Sponsors: ₹2,00,000 each*
- *Silver Sponsors: ₹1,00,000 each*
- *Bronze Sponsors: ₹50,000 each*